

Kowndinya Boyalakuntla

Ph.D. student in Computer Science | Rutgers University

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Research interests

I work on learning and planning for robot manipulation and whole-body control. My current research explores learned world-models and how to turn human motion capture into useful action abstractions for long-horizon planning in humanoid loco-manipulation.

Education

Rutgers University

Ph.D. in Computer Science

Advisor: Prof. Abdeslam Boularias

Sep 2024 – present

Rutgers University

M.S. in Computer Science; GPA: 3.8/4.0

Advisor: Prof. Abdeslam Boularias

Thesis: Context-Aware Entity Grounding with Open-Vocabulary 3D Scene Graphs.

2022 – 2024

Indian Institute of Technology Tirupati

B.Tech. in Computer Science and Engineering; GPA: 8.42/10.0

Advisors: Prof. Sridhar Chimalakonda, Prof. Kalidas Yeturu

Jul 2017 – Jul 2021

Manuscripts under review

- PL-MPC: Beyond Policy Alignment: Closing the Planning–Learning Loop for Robot Control with Learned World Models**
Kowndinya Boyalakuntla, Yuhan Liu, Abdeslam Boularias
Manuscript, 2026. Under review at ICRA 2027. [\[Paper\]](#) [\[Project\]](#) [\[Code\]](#)
- TRACE: Plan-Conditioned Imitation for Robust Object Retrieval under Self-Occlusion in Dense Clutter**
Kowndinya Boyalakuntla, Ajinkya Pawar, Abdeslam Boularias, Jingjin Yu
Manuscript, 2026. Under review at ICRA 2027. [\[Paper\]](#) [\[Project\]](#) [\[Code\]](#) [\[Data\]](#)

Research experience

Rutgers University

Graduate Student Researcher · New Brunswick, NJ

Research in robot learning, visual perception, and manipulation with Prof. Abdeslam Boularias. Work includes scene grounding, language-guided rearrangement, dynamic grasping, learned world models, and manipulation under self-occlusion.

Jan 2023 – present

RISHA Lab, IIT Tirupati

Undergraduate Student Researcher · Tirupati, India

Research in empirical software engineering and human–computer interaction, including tools for evaluating software repositories and generating energy-aware mobile applications.

Dec 2019 – May 2022

University of Waterloo

Visiting Scholar · Waterloo, Canada

Improved RepoReaper for large-scale GitHub repository analysis, increasing recall and reducing false positives without relying on GHTorrent.

Jul 2020 – Sep 2020

Teaching

Instructor, Rutgers University

CS440: Introduction to Artificial Intelligence Undergraduate course

Fall 2026

Peer-reviewed publications

1. **KARL: Kalman-Filter Assisted Reinforcement Learner for Dynamic Object Tracking and Grasping**
Kowndinya Boyalakuntla, Abdeslam Boularias, Jingjin Yu
IROS, 2025 [\[Paper\]](#) [\[Code\]](#)
2. **DAP: Diffusion-based Affordance Prediction for Multi-modality Storage**
Haonan Chang, Kowndinya Boyalakuntla, Yuhan Liu, Xinyu Zhang, Liam Schramm, Abdeslam Boularias
IROS, 2024 [\[Paper\]](#) [\[Code\]](#)
3. **LGMCTS: Language-Guided Monte-Carlo Tree Search for Executable Semantic Object Rearrangement**
Haonan Chang, Kai Gao, Kowndinya Boyalakuntla, Alex Lee, Baichuan Huang, Harish Udhaya Kumar, Jingjin Yu, Abdeslam Boularias
IROS, 2024 [\[Paper\]](#) [\[Project\]](#) [\[Code\]](#)
4. **Context-Aware Entity Grounding with Open-Vocabulary 3D Scene Graphs**
Haonan Chang, Kowndinya Boyalakuntla, Shiyang Lu, Siwei Cai, Eric Pu Jing, Shreesh Keskar, Shijie Geng, Adeeb Abbas, Lifeng Zhou, Kostas Bekris, Abdeslam Boularias
CoRL, 2023 [\[Paper\]](#) [\[Project\]](#) [\[Code\]](#)
5. **eGEN: an energy-saving modeling language and code generator for location-sensing of mobile apps**
Kowndinya Boyalakuntla, Marimuthu Chinnakali, Sridhar Chimalakonda, Chandrasekaran Kalicheti
ESEC/FSE, 2022 [\[Paper\]](#) [\[Code\]](#)
6. **RepoQuester: A Tool Towards Evaluating GitHub Projects**
Kowndinya Boyalakuntla, Meiyappan Nagappan, Sridhar Chimalakonda, Nuthan Munaiah
ICSME, 2022 [\[Paper\]](#) [\[Code\]](#)
7. **GitQ-towards using badges as visual cues for GitHub projects**
Akhila Sri Manasa Venigalla, Kowndinya Boyalakuntla, Sridhar Chimalakonda
ICPC, 2022 [\[Paper\]](#) [\[Code\]](#)
8. **A Platform for Large Scale Auto Annotation of Scanned Documents Featuring Real-Time Model Building and Model Pooling**
Prashanth Komuravelli, Kowndinya Boyalakuntla, Vijay Chilaka, Teja Dande, Vidya Rodge, Ramya Velaga, Reena Abasaheb Deshmukh, Kalidas Yeturu
CVIP, 2021 [\[Paper\]](#) [\[Code\]](#)

Honors & awards

2022 Project showcase, IInvenTiv All IITs R&D Fair. Research project advised by Prof. Kalidas Yeturu, showcased at IIT Delhi.

2019 Gold Medal, Inter IIT Technical Meet 8.0. Ashoka's Tech for Change Challenge at IIT Roorkee; high-preparation event among 20 IITs.

Leadership & mentoring

Google Developer Student Clubs, IIT Tirupati

Feb 2019 – Jul 2020

Community Lead

Organized 12 workshops and hackathons for more than 70 students; led 15 mentoring sessions on Flutter and Android.

Technical skills

Programming: Python, C++

Robotics and learning: PyTorch, TensorFlow, OpenCV, PyBullet, Isaac Gym, ROS, Kornia

Visualization: Three.js

Professional service & activities

Reviewer: IROS (2025, 2026); CoRL (2026); IJRR (2026); ICAPS (2026).

2021 Developed an Android application for IIT Tirupati's convocation.

2020 Completed 80 hours of National Service Scheme service; developed the IIT Tirupati NSS website and led its maintenance team.